



Computational Models Trained on Deep Mutational Scanning Data to Predict and Design Functional Transcriptional Effectors

*Raeline Valbuena¹, *Akshat Nigam², Josh Tycko¹, Kaitlyn Spees¹, Aradhana¹, Sophia Arana¹, Anshul Kundaje^{1,2}, Lacramioara Bintu³, Michael Bassik¹
Stanford Department of Genetics¹, Stanford Department of Computer Science², Stanford Department of Bioengineering³

ABSTRACT

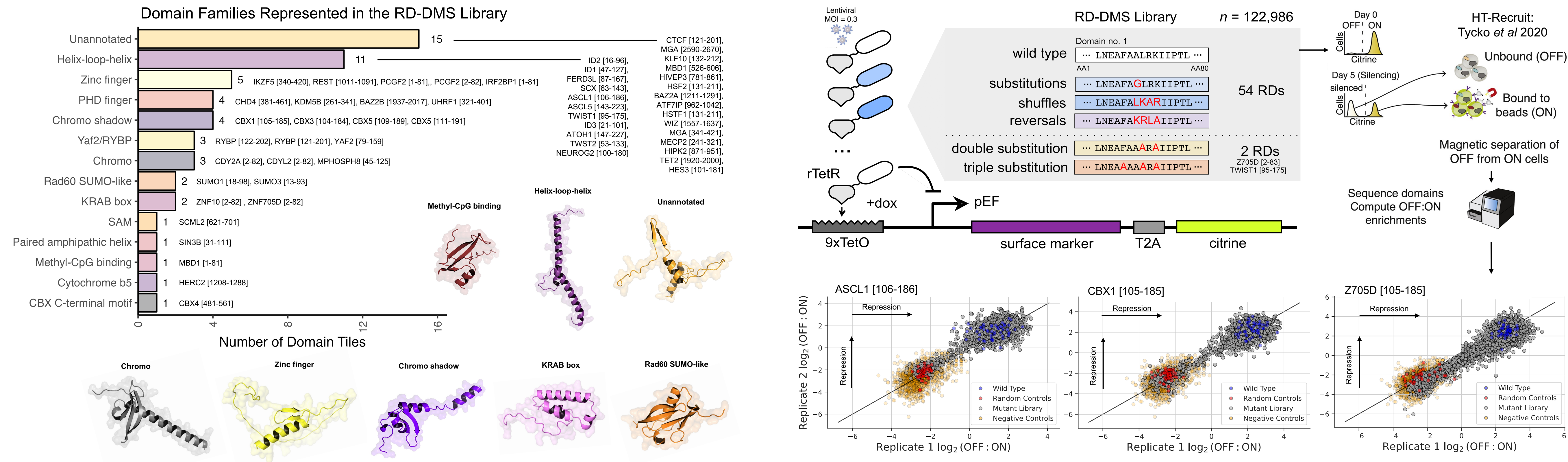
Protein structure and sequence predictions have become increasingly accurate as diverse types of biological knowledge are paired with improved deep learning algorithms. The ability to accurately predict protein function in a similar manner would greatly facilitate the development of impactful tools for bioengineering, especially in the field of gene regulation. **Here we present our work to measure over 115K phenotypes for repressor domains (RDs) with amino-acid resolution, and to build integrative computational models to predict and characterize repressor function.** Using High-Throughput Recruit¹, a pooled assay that measures the transcriptional effects of hundreds-of-thousands of proteins in parallel, we generated quantitative deep mutational scanning data for 54 RDs sampled from diverse effector families.

Then, to begin predicting the molecular phenotypes of the RDs, we designed a computational workflow to estimate end-state free energy calculations for RD-cofactor interactions. We find a moderate correlation between the repressive strength of some mutants and the interaction energy with their respective binding partners, suggesting that our pipeline may be used to predict functionally-important cofactor interactions, in some cases. Additionally, we present our current progress in developing a deep learning approach to predict functional repressive strengths of novel effector domains. Ultimately, we aim to build a deep learning model proficient in the functional prediction of clinical variants and inverse design of fine-tuned *de novo* synthetic effectors.

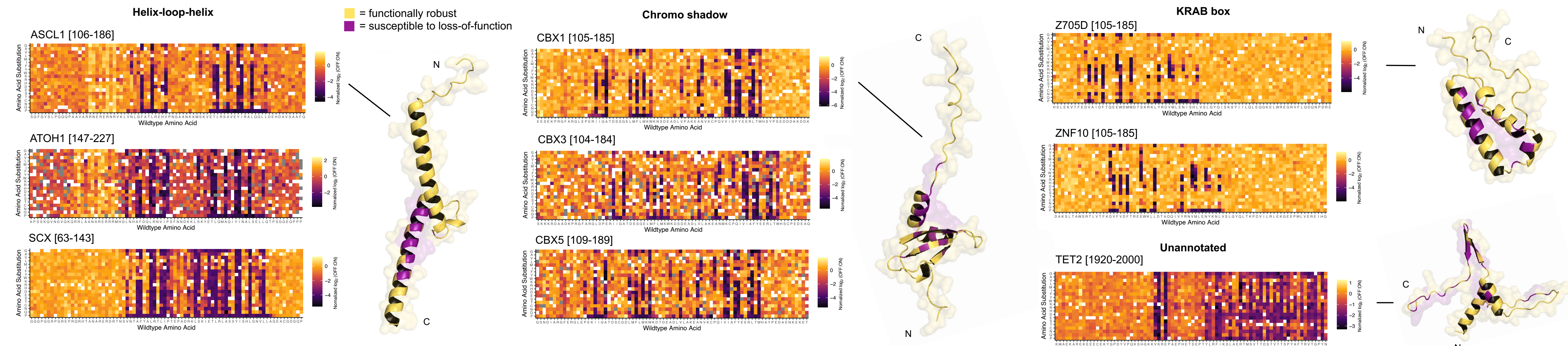
References:

1. Tycko, J. *et al.* High-Throughput Discovery and Characterization of Human Transcriptional Effectors. *Cell* 183, 2020-2035 (2020).
2. Evans, R. *et al.* Protein Complex Prediction with AlphaFold-Multimer. *bioRxiv* 2021.10.04.463034 (2021).
3. Miller, B. III. *et al.* MMPBSA.py: An Efficient Program for End-State Free Energy Calculations. *J. Chem. Theory Comput.* 8, 9, 3314–3321 (2012).
4. Lin, Z. *et al.* Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* 379, 1123-1130 (2023).

MEASURING REPRESSOR FUNCTION OF MUTANT LIBRARIES WITH HT-RECRUIT

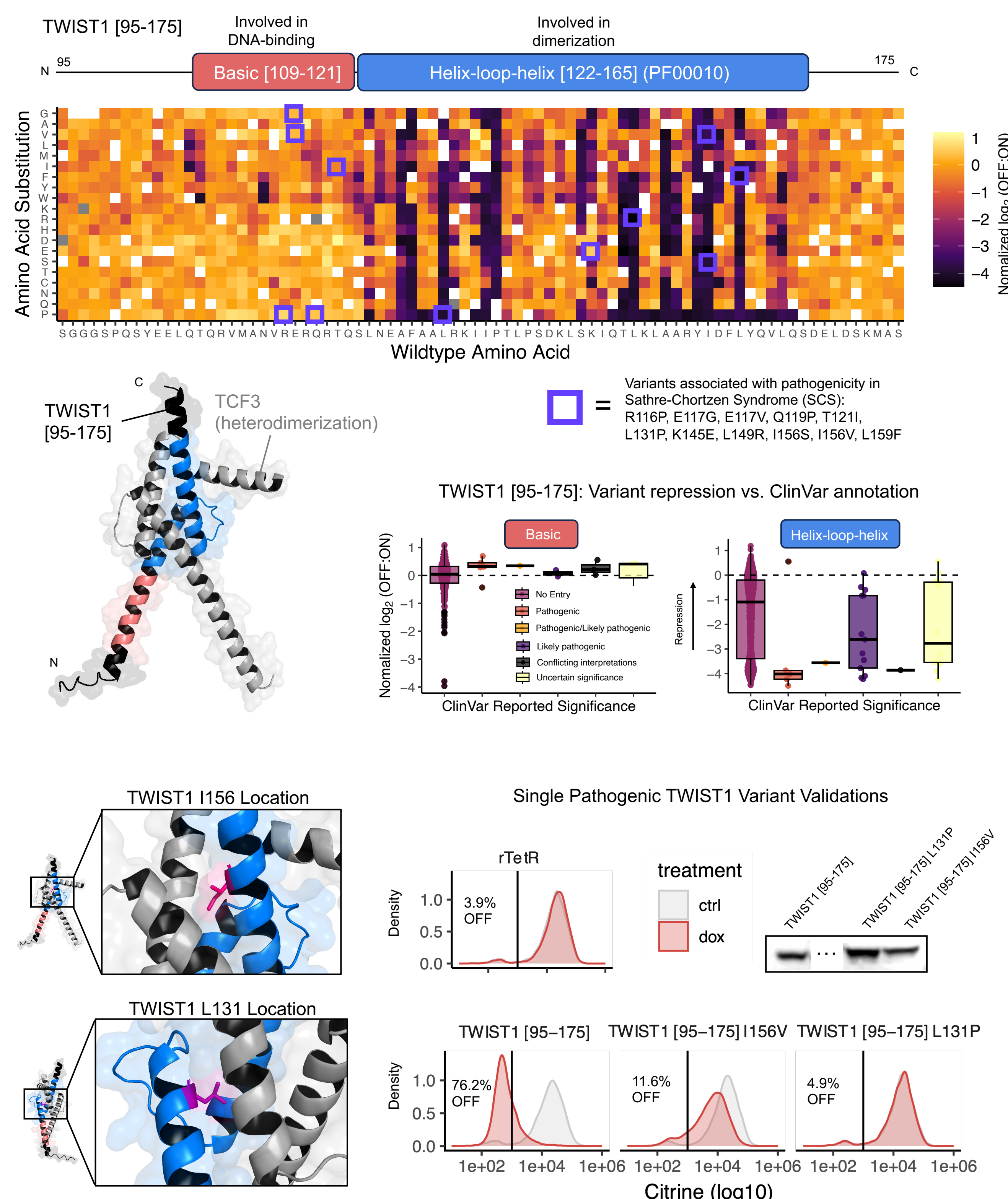


Single substitutions reveal functionally-relevant structures in domain architectures



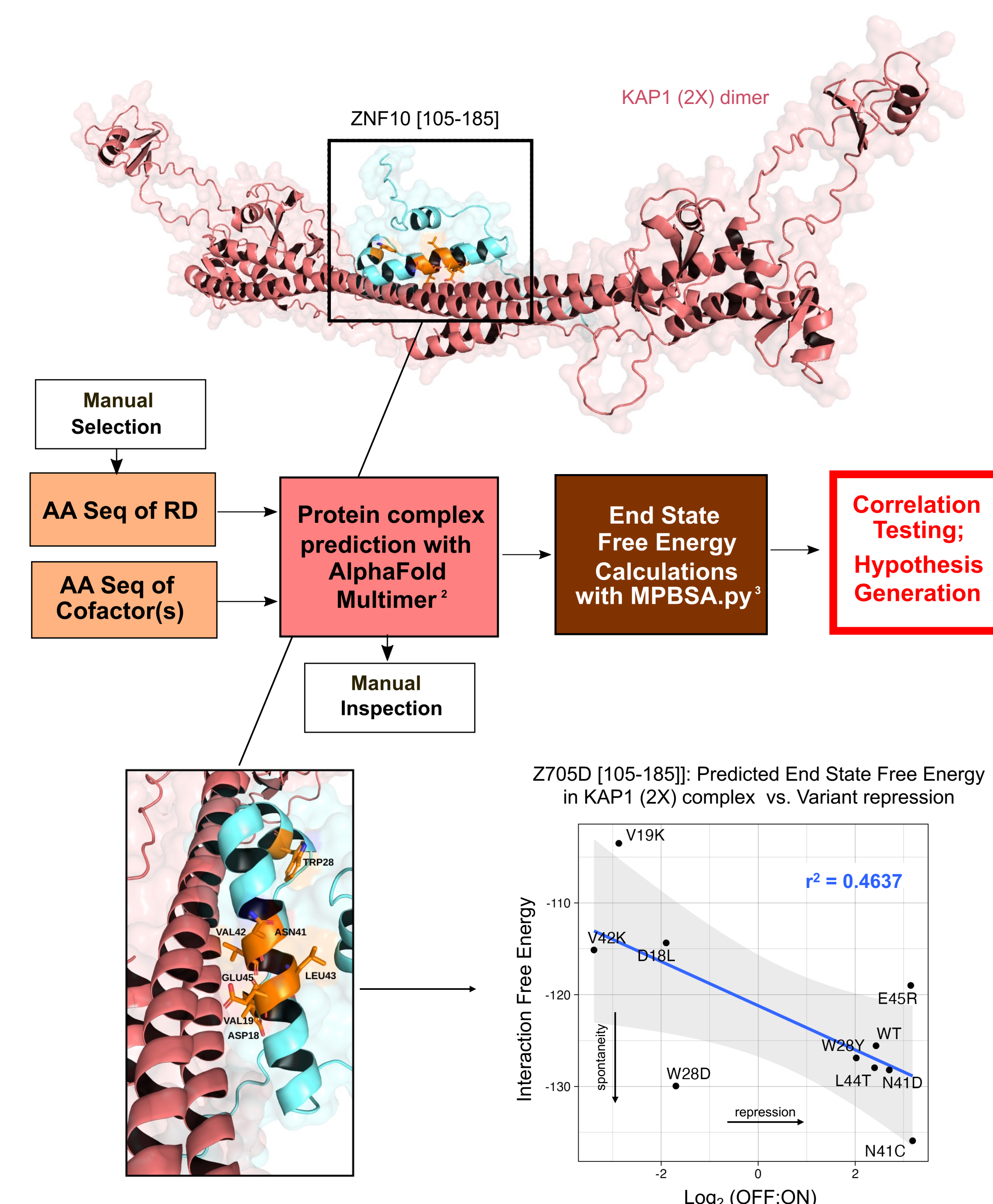
UNDERSTANDING CLINICAL VARIANTS

TWIST1 variants associated with Saethre-Chotzen syndrome decrease its repressive function



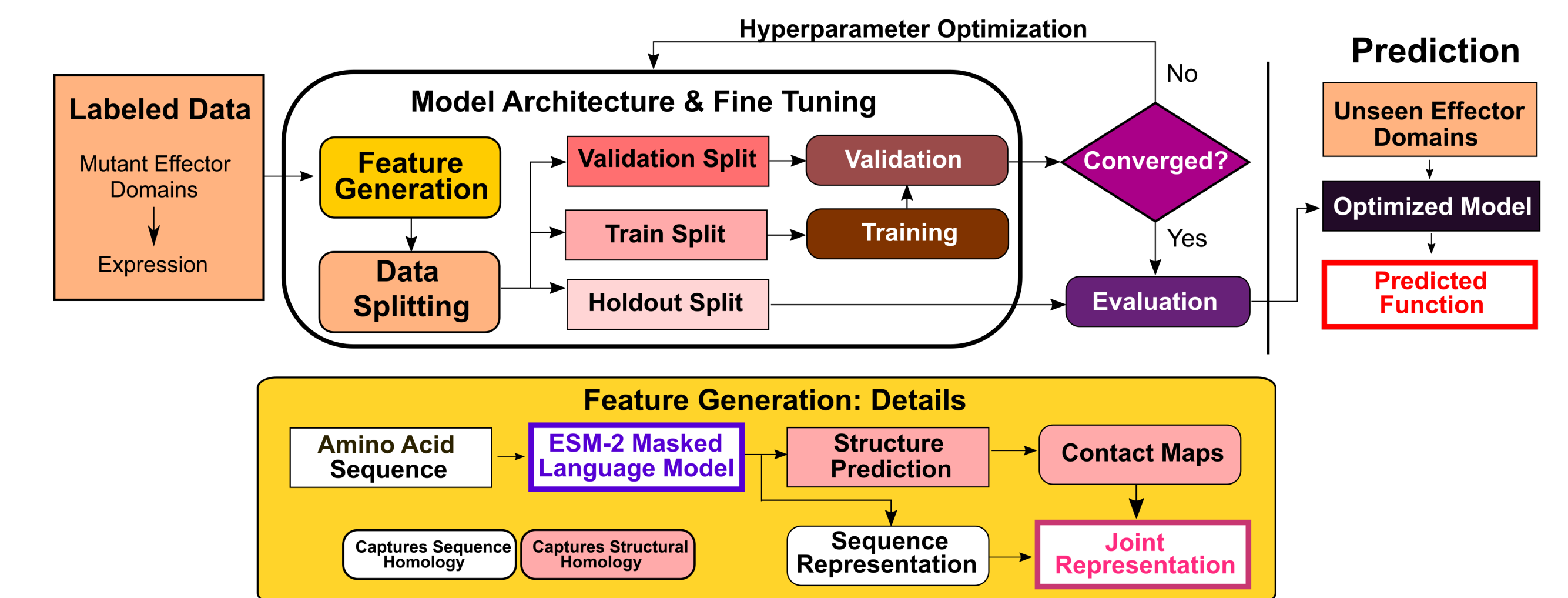
ASSESSING MOLECULAR FUNCTION

KRAB box variants along the KAP1 binding interface decreased predicted binding energy



GENERATING A NEURAL NETWORK TO PREDICT REPRESSOR FUNCTION

Calibrating the large transformer protein language model ESMFold⁴ (ESM2-b) with 650 million parameters



FUTURE DIRECTIONS

Future work will test the performance of our deep learning model: we will generate functional predictions for unseen RD domains and experimentally quantify their repressive activity in our recruitment assay to evaluate model performance. These data will be used to refine our model and its ability to predict repressor function. In combination with pipelines assessing molecular functions (e.g. cofactor interaction strengths), we aim to build a robust computational framework for modeling repression.

ACKNOWLEDGMENTS

Thank you to all members of the Bassik, Bintu, and Kundaje labs for their continual support and feedback. Various unpublished cell lines, libraries, plasmids and reagents are gifts from Aradhana, David Yao, Kaitlyn Spees, Mike Vi Van, and Nicole Del Rosso.

